# Numerical Optimization - Challenge Problems

Thursday, September 28, 2017 9:04 AM

$$\max_{\{w_A, w_G\}} \quad \text{Sharpe} = \frac{R_p - R_f}{\sigma_p}$$

$w_A = 34\%$

$w_A \geq 0$ $\quad w_G \geq 0$

$w_A \leq 1$ $\quad w_G \leq 1$

$w_A + w_G = 1$

$w_A \geq 0$ Slack

$34\% \geq 0$ Slack

$\downarrow$

$36\% \geq 0$

Max Sharpe = 0.46 Optimal Feasible

Feasible $w_A = 50\%$ Sharpe = 0.45

$w_A = 51\%$

=

$w_A = 0\%$

$w_G = 100\%$

$0\% = w_A \geq 0$ Binding

$0\% = w_A \leq 1$ Slack

$100\% = w_G \geq 0$ Slack

$100\% = w_G \leq 1$ Binding

$w_A + w_G = 1$ Binding

$$\max \quad \text{Sharpe} - 4 \cdot \text{Skew}$$

Objective $\boxed{\dfrac{R_p - R_f}{\sigma_p} - 4 \cdot S_p}$

$100

$w_1 = 40\% \Rightarrow \$40$

$w_2 = 50\% \Rightarrow \$50$

$w_3 = 10\% \Rightarrow \$10$

$40/120 = 33\%$

$\$50/120$

$10/12$

+$20

max sharpe